

Prologue

A fine gold inlay is the best insurance against recurrent decay.

I found growing up in the dental lab exciting. I was interested in what I saw my dad doing at his bench from a young age, maybe 8 or 9. I remember being a little sad as it expanded throughout the basement and took over our small roller-skating area — from playground to a work area, a child's perceived loss, but really not. The first effects of OSHA, I guess. The basement slowly filled up with benches. I remember helping my dad with floor plans, cutting out small construction paper benches to scale so that we could maximize space and improve workflow.

I've often thought about those early years. I'm sure I was bothersome from time to time, looming over from the stairwell above his bench, or standing right next to him asking questions. But Dad just always had a technical explanation, and my brain soaked it up like a sponge. If I bothered him, he never let it show. From there Dad began my training, with terminology and tools, maintaining equipment and cleaning the areas. Then into the technical at about 11 or 12 years old, starting in the partial dentures department.

I remember OSHA conversations, and Dad was always a stickler for following instructions, including reading prescriptions. And he was always vigilant about safety in the lab. I remember him traveling to meetings more frequently while he was president of the Pennsylvania Association of Dental Laboratories (PADL). I remember the binders he had with all his information very organized. I remember his drive to bring standards to the profession from the laboratory side. I was still a little bit too young to comprehend the gravity of what he was promoting. I understand today.

Dental technology was a perfect match for my nature. I fell into it, and it fit like a glove. The hands were quick to learn, and it seemed to come fairly easy to me. After graduating high school and serving in the Air Force for a while, I returned to the lab and began my technician career in earnest. Over the next two decades or so, after Dad closed his lab due to economic pressures in the 80s, I

worked at several different labs — some in-office, some small, some big, one with a trial partnership — then I opened my own lab. After nine years and life and dentistry changes, I closed my lab and joined my brother Don to help him grow his business.

He owned Atlantic Precious Metal Refining, serving labs around the east to Midwest of the country. He wanted to expand, and expand he did. Starting at the smelting and sales positions and branching out to the south and west, I continued to observe the quantity of precious metals in dentistry start its slow decline. By the time I left, the company was much bigger, serving the entire country, and leading in the dental market for scrap refining. From there I moved on to a large local lab and expanded their implant department as well as developed in-lab continuing education for their technicians. My absence from the bench while refining scrap had me entering into the world of digital right around the time that Scott was bringing milled gold to the marketplace. And a few years later, I saw that the new technology was being underused by the community, starting with the dentists. They were still mostly unaware of milled gold. Many are just now realizing its benefits.

Scott and I had a meeting, and although it was a daunting challenge — to shift the perspective of the dental community — I accepted, maybe a bit foolheartedly.

I saw a problem in dentistry that nobody was paying attention to, and I saw that this new technology improved upon the old technology by an order of magnitude. This is valuable information for dentists, labs, and all patients. Because this technology changes the landscape for initial treatment of caries presenting. And caries is not only the number one disease in the world, but its occurrence rate is on a 30-year climb. And so is periodontal disease and TMD. That shouldn't be, if dentistry is successful. All the modern implant technology is great, if the patient needs it. But we'd lost focus as a profession on the mother wound of restorative dentistry: filling a cavity so that it remains filled, protective, and long-lasting.

And that's what landed me here, building Easy Gold Crowns — making the final connection back to the dentist and patient, now with milled gold. Easy Gold Crowns launched in 2019 and I've spent the last seven years refining the digital gold design and polishing workflow, and offering this advancement to the dentist community. I recently launched an app, Gold Dentistry Network, to connect patients seeking gold to gold-delivering dentists, and gold-delivering dentists with gold-

producing labs — whether milled or cast. All are welcome. Any gold on a posterior tooth is a win. Some dentists still prefer cast, and some labs still have skilled technicians who can produce it.

This book is what I found when I followed that wound back to its source — through the codes, the curriculum, the insurance tables, the classroom — and what it will take to heal it.

Part 1: The Wound

Chapter 1: The First Cavity

We have a problem. The number one disease in the world is caries. In fact, the rate of occurrence has gone up over the past thirty years, and recurrent decay is the number one reason for restoration failure. What we are doing — how we are treating caries — isn't working for patients.

This is our primary reason for existence as a profession: fighting decay and preserving teeth. We adopted an oath to do that. Our ethics say we have a duty to promote excellent oral health care treatments.

But somewhere along the way, the profession lost focus. All the modern implant technology is great, if the patient needs it. But we've drifted from the mother wound of restorative dentistry: filling a cavity so that it remains filled, protective, and long-lasting.

Think about what a first cavity represents. A tooth that was whole. A patient who trusted us. And a choice of material that will determine whether that tooth is still intact in ten years, or thirty, or fifty.

Most cavities present on posterior teeth — outside the smile zone, in areas difficult to clean and difficult to work on. Molars and premolars do the grinding work of a lifetime. Harmonious function with the opposing tooth is the number one priority for preserving them. And dental teams are the only people who see these teeth on a regular basis. The patient can't monitor them. They trust us.

Chapter 2: The Composite Compromise

How are we treating that first cavity? The most common first treatment of a restorable cavity is a light-cured composite direct filling. It is convenient when a patient presents. But the convenience is more temporary than the composite.

The material degrades in the mouth. Over a few years of chewing forces, temperature swings, and pH plunges, the margin — the interface where the filling meets the tooth — begins to break down. Composite shrinks as it cures, and final cure takes 24 to 48 hours. That shrinkage creates internal stress toward center mass, which can develop into a microscopic gap. Bacteria don't need a big door. A crack is an invitation.

Eventually decay returns. The filling is replaced. More tooth structure is removed. The cavity gets larger. The next filling is larger. The cycle repeats.

This ends up costing everybody more, especially the patient. They lose tooth structure each time decay returns. They lose time. They lose the option of a small, conservative repair, because with each replacement the tooth becomes less repairable. The cascade is predictable: filling, then larger filling, then crown, then root canal, then extraction, then implant. All of it downstream of that first material choice.

And dentists know it. Every day in operatories across the country, they wrestle with the same frustrations:

The polymerization shrinkage that pulls the filling away from the cavity walls as it cures. The post-operative sensitivity that follows — was it the bond, the cure, the bite? The layering: oblique increments, one wall at a time, each cured before the next can be placed. The rubber dam. The isolation. One drop of saliva or blood and the bond is compromised. The proximal contact that's never quite right — too light, too tight, or open entirely. The finishing: cup, disc, point, paste — more steps than the placement itself. The shade that looked perfect on the guide and mediocre in the mouth. The patient who calls back three days later: "It's still sensitive." The bite adjustment. The second visit. The margin that stains brown at the two-year hygiene check. The radiograph at year five that shows a shadow beneath the filling — recurrent decay, and it all starts again.

These are not isolated complaints. They are the same frustrations, voiced by every dentist, in every practice, because they are inherent to the material. Composite asks the operator to overcome its limitations through technique. And when the technique fails — as it eventually must, given enough time in the mouth — the patient pays the price.

Composite is the least durable material that we use. This is not opinion. It is not a debate within the profession. The literature is clear.

Chapter 3: The Gold We Left Behind

The most successful material for filling cavities is gold. There is no argument about this among dentists who have placed both. Gold wears at the same rate as natural teeth. It does not shrink. It does not degrade. It seals the margin. These are not optional features that depend on technique or conditions. They are inherent to the material — delivered by the metal itself, asking nothing of the operator beyond competent placement.

Everyone knows gold has intrinsic value. That's the problem. The price per ounce, the commodity ticker, the sense that gold is expensive before it is anything else — this is what the insurance industry sees, what patients hear, and what the profession has internalized. But intrinsic value is the wrong measure. What matters is inherent value: the properties gold possesses that no other restorative material can match. Strength without brittleness. Malleability without weakness. A wear rate that matches enamel. A margin that stays closed. These are what deliver durability, longevity, and the silent decades of service that follow.

The profession focused on what gold costs and forgot what gold does.

But increasing gold costs back in the 1970s drove dentistry to seek out substitutes. This was a practical challenge — gold had become expensive and volatile. Labs had to adjust. Insurance companies had to adjust. And in that adjustment, the profession began a slow pivot away from its best material, not because something better had arrived, but because something cheaper and more predictable had.

Old gold devices — sixty, seventy, even eighty years old — are still serving patients today. Most of these gold success stories go untold and unnoticed to the grave. Which, in a way, is how it should be. A restoration that never fails never makes news. The patient who received a gold inlay at thirty and died with it still intact at eighty never had a reason to think about it again.

But that silence has a cost. Because the profession stopped talking about gold, too.

Why is it these days that dentistry is the only place gold isn't pretty? How did this idea come about, given its superior performance?

Chapter 4: The Other White Material

There is a secondary problem concerning the other class of white materials we use: ceramics. These materials are two to four times harder than natural teeth. They do not wear down like teeth do. For the sixty percent of the population with some degree of malocclusion, ceramics are actually contraindicated — they abrade the opposing dentition over time.

So the profession's answer to composite's failure has been to offer a material that eventually destroys the opposing tooth over time. They do not wear harmoniously like natural teeth or gold. The patient who receives a ceramic crown at forty may be back at sixty with a worn-down opposing molar, and the cascade begins again, now on two teeth instead of one.

But there is a deeper problem with ceramics, and it lives at the margin — the very place the ceramic restoration is supposed to win.

The cement that bonds a ceramic crown to the tooth is a resin composite. And resin composites are vulnerable in a way that few dentists consider: they are biodegradable. Not over decades. Actively. Enzymatically.

In 2018, Huang and colleagues at the University of Toronto identified a specific esterase enzyme produced by *Streptococcus mutans* — the primary caries-causing bacterium — that hydrolyzes methacrylate-based resin monomers. The enzyme, SMU_118c, breaks down bisGMA and TEGDMA, the universal building blocks of composite fillings and resin cements. It retains sixty

percent of its activity after three weeks. It functions at neutral pH and at the acidic pH of active caries. Its confirmed effect is to compromise the restoration-tooth interface, increasing bacterial microleakage.

The same research group had already demonstrated in 2010 that this biodegradation directly increases bacterial infiltration at the resin-dentin interface. The bacteria that cause caries do not merely exploit passive gaps in a failing restoration. They actively create those gaps by consuming the cement that seals them.

This means the ceramic crown does not, in fact, succeed at the margin. Its cement does exactly what composite fillings do: it degrades under bacterial attack, opens a microscopic pathway, and invites recurrent decay to begin its silent advance. The patient sees a beautiful white crown. *S. mutans* sees lunch.

Neither white material solves the problem it was asked to solve. Composite fails at the margin through shrinkage. Ceramic punishes the opposing arch through hardness. And both depend on resin cements that the oral bacteria are equipped to consume.

And all the while, the gold devices placed before most practicing dentists were born are still in service, wearing gently alongside the teeth they protect. Gold doesn't bond. It seals. There is no resin layer to degrade, no cement interface for bacteria to colonize and consume. The margin is metal against tooth, and the bacteria have nothing to eat.

Part 2: The Cascade

Chapter 5: The Gold Window Closes

The story of how dentistry lost gold begins, strangely, in Washington.

In 1971, President Nixon closed the gold window. Overnight, the dollar was no longer convertible to gold at a fixed rate. Gold became a freely traded commodity, and its price — stable at \$35 an ounce for decades — began a long, volatile climb. Day traders and speculators moved in. The metal that dental labs had relied on for a century, priced predictably and worked reliably, became subject to the whims of global markets.

Insurance companies noticed. They build their business on predictability — coded procedures, fixed reimbursements, actuarial tables. A restoration material whose cost could swing twenty percent between a treatment plan and a seating appointment was a problem. They needed something stable. Something cheap. Something predictable.

The petrochemical industry had an answer. Composite resins — petroleum-based, mass-produced, and priced like a manufactured product rather than a traded metal — fit the insurance model. The cost per gram was stable. The production could scale. And the material's shorter lifespan, whatever its clinical drawbacks, was not a liability in a reimbursement system that bills per procedure.

What happened was the first subtle, and in some ways more dangerous, erosion of the standard of care: the economic incentives of the largest payer in dentistry aligned with the inferior product, and the profession adjusted around the money.

The insurance industry had measured gold's intrinsic value — the price per ounce — and found it inconvenient. It never measured gold's inherent value — the margin that stays closed, the decades

of silent service — because inherent value does not appear on an actuarial table. The system was built to price what could be counted and to ignore what could not.

Chapter 6: What the Instructors Knew

By 1986, the shift was well underway. But the people training the next generation of dentists knew what was happening.

I learned this from a conversation with a colleague who was in dental school that year. Students were being told something remarkable — and remarkably revealing. The instruction went like this: if a patient presented with amalgam fillings and requested their removal because they contained mercury, replacing them could cost you your license. But if the same patient requested the same procedure for cosmetic reasons, and the patient accepted the risk, it was permissible to replace them with composite.

Read that again. The instructors were teaching that composite was a risk the patient had to accept. They knew. They knew the new material was a downgrade in protection. They knew amalgam was more durable. And they built an ethical framework around managing the narrative: don't remove amalgam for health concerns — that's dangerous to your career. Remove it for cosmetics — that's the patient's choice, and their risk to bear.

The clinical reality was not in dispute. The instructors had no illusions that composite was better. What they taught was how to navigate the growing gap between what was best for the patient and what the system would support.

Chapter 7: The Codes That Weren't Written

The reimbursement system in American dentistry runs on codes. Every procedure a dentist performs has a CDT code — a number that tells the insurance company what was done and what it should pay. If a procedure has no code, it effectively does not exist in the insurance framework.

The dentist can perform it. The patient can pay for it out of pocket. But the vast machinery of dental benefits — the system that determines what most patients can afford and what most dentists will offer — will not touch it.

There is no CDT code for a gold inlay.

There are codes for composite fillings. Codes for amalgam fillings. Codes for ceramic inlays. Codes for gold crowns. But a gold inlay — the most conservative, most durable restoration for a small to medium cavity on a posterior tooth — has no code. The absence is not an oversight. It is a structural decision that steers treatment toward reimbursable materials and away from the one that works best.

When there is no code, insurance does not reimburse. When insurance does not reimburse, most patients cannot afford the treatment. When most patients cannot afford the treatment, most dentists stop offering it. When most dentists stop offering it, dental schools stop teaching it. When dental schools stop teaching it, a generation of dentists graduates having never placed one.

This is how a material disappears. Not by being banned. Not by being proven inferior. But by being coded out of existence.

Chapter 8: The Curriculum Gap

By the late 1990s, gold had all but vanished from dental school curricula. A student entering dental school in 1998 would graduate in 2002 having never prepared a tooth for a gold inlay, never taken an impression for one, never seated one, never polished one. The faculty who knew how had retired or been reassigned. The lab support had moved to ceramics and composites. The textbooks mentioned gold as a historical footnote.

Today, a dentist in their mid-forties — twenty years into practice, at the peak of their clinical skill — may have never seen a gold inlay placed. Not as a student. Not as a resident. Not in their own practice. They are not opposed to gold. They are not hostile to it. They simply do not know it exists as a practical option.

This is not their failure. It is a failure of the institutions that trained them.

A profession that takes an oath to preserve teeth has, over four decades, systematically removed from its practitioners' hands the material that does it best. Each step in the chain felt reasonable at the time. The cost of gold was rising. The insurance codes were complicated. The curriculum was crowded. The new materials were improving. No one decision was catastrophic.

But the cumulative effect is catastrophic. The world's number one disease is treated, in most mouths, with its least durable material. And the people responsible for fixing it — the dentists — were never shown the alternative.

Part 3: The Alternative

Chapter 9: The Margin

Every restoration survives or fails at the margin. Recurrent decay is the number one reason given by the ADA for all restoration failure. The interface where the filling or inlay meets the tooth is where the war is won or lost. The better we seal it, the less likely decay will return.

No matter how refined the technique, every restoration has a clinically acceptable measurable gap at the margin. We rely on our cements to seal that gap. Old school zinc phosphate cement has a long track record of success when handled properly — but it is also notoriously technique-sensitive. Mixed too thin, it weakens. Mixed too thick, it fails to flow. The window for a perfect mix is narrow, and few dentists today use it regularly enough to stay in that window.

Modern luting agents are, in fact, specialized composite materials. Many contain glass ionomer, a known anti-cariogenic that releases fluoride over time — an appealing feature. But being composite resins at their core, these cements are subject to the same deterioration as any composite material in the oral environment. They degrade. They wear. They lose their seal.

And this is where the size of the gap matters profoundly.

The wider the marginal gap, the more cement surface area is exposed to the hostile oral environment — acid attacks, temperature swings, bacterial colonization. The greater the exposed surface area, the faster the cement deteriorates. The faster the cement deteriorates, the sooner the seal is compromised. And the sooner the seal is compromised, the sooner recurrent decay begins its advance beneath the restoration, invisible to the patient, invisible to the dentist, until the next radiograph or the next symptom.

Composite shrinks as it cures, and final cure takes 24 to 48 hours. That shrinkage creates internal stress toward center mass, which can develop into a microscopic gap.

Ceramic restorations, though they do not shrink, are milled or pressed to fit — and every milling bur and pressing die leaves a gap that must be filled by cement. When ceramics first entered widespread use, dentists designed them with tight margins — and watched them fracture. The early milling machines and sintering processes were imprecise, leaving microscopic pressure points on the intaglio surface. Upon seating, those pressure points became fracture points.

The profession's workaround was to increase the cement gap. A lot. The gap went from the cement manufacturers' recommended 30 microns to over 70 microns — a 230 percent increase. The fractures stopped. But two new problems began.

First, the wider gap exceeded the cement's design specifications, causing restorations to de-cement through shear failure — the cement layer was simply too thick to hold. Second, crowns began to feel loose at try-in, lacking the resistance and retention that a tight fit provides. As one dentist put it to me: like socks on a rooster.

The profession adapted to this degradation and adopted it as the new standard of care. Today, a 70-micron cement gap on a ceramic restoration is considered normal. The cement manufacturers' 30-micron recommendation is ignored. The loose fit is accepted. The de-cementation risk is managed with ever-stronger bonding agents — which are themselves composite materials subject to the same in-vivo deterioration we have already discussed.

This is what an erosion of the standard of care looks like up close. Not a decision to do worse. An adaptation to a material's limitations, repeated until the adaptation became the standard, and the standard became invisible.

And now we know something else about that 70-micron gap that was not understood when the profession made its peace with it. In 2018, researchers at the University of Toronto identified a specific esterase enzyme produced by *Streptococcus mutans* — the primary caries-causing bacterium — that actively hydrolyzes methacrylate-based resin monomers. The enzyme breaks down bisGMA and TEGDMA, the universal building blocks of resin cements. It retains sixty percent of its activity after three weeks. It functions at neutral pH and at the acidic pH of active

caries. The bacteria that cause decay do not merely colonize a gap — they consume the cement that fills it.

Every additional micron of cement gap is additional surface area of digestible substrate exposed to bacterial esterase activity. A 70-micron gap is more than twice the cement surface area of a 30-micron gap. Twice the food. Twice the degradation rate. Twice the biological vulnerability.

The 230 percent increase in cement gap was adopted to solve a mechanical problem: ceramic fractures under seating pressure. But it created a biological problem: it set a larger table for the bacteria. Size matters. And the size that was chosen — 70 microns — happens to be the size that maximizes the meal.

Gold never required that trade-off. It achieves 30 microns or less — within the cement manufacturer's specification — because gold does not fracture under seating pressure, does not require a relief gap, and does not ask the cement to compensate for the material's weaknesses. These are inherent properties, not operator achievements. The dentist does not overcome gold's limitations through technique — gold has no limitations to overcome. The margin gap on a well-designed gold inlay is what the cement was engineered for.

This is not theory. The literature is unambiguous. Gold restorations demonstrate the lowest rates of recurrent decay of any restorative material. The reason is not complicated. The margin stays closed.

Chapter 10: The Evidence

In 2009, Bandlish and Mariatos published a retrospective study of cast gold onlays placed in a general dental practice. The results: ninety-seven percent survived thirty years or longer. Not ninety-seven percent with repairs. Not ninety-seven percent with monitoring. Ninety-seven percent still in service, still protecting the tooth, still sealing the margin, three decades after placement.

Compare this to the composite restoration placed today. The average lifespan of a composite filling is seven to ten years. That means the patient who receives composite for their first cavity can

expect two, three, or four replacements over the same thirty-year window that the gold patient never thinks about their tooth again. And each replacement removes more healthy tooth structure — the cavity grows, the tooth weakens, and the cascade toward crown, root canal, and extraction accelerates.

The math is uncomfortable. A profession that prides itself on prevention and conservation has adopted as its default a material that guarantees more intervention.

Chapter 11: Milled, Not Cast

The old objection to gold was practical. Cast gold restorations required a wax-up, an investment mold, a casting centrifuge, and a skilled technician whose hands could compensate for the inherent imprecision of the process. It was slow. It was expensive. It was technique-sensitive. And the results, though excellent in the right hands, varied with the technician.

Direct-milled gold changes every part of that equation.

A digital scan of the prepared tooth feeds into CAD software where the restoration is designed to micron-level precision. The design is milled directly from a solid gold blank — no wax, no mold, no casting variables. The result is a restoration that reproduces the design exactly. Johnson and colleagues demonstrated in 2017 that milled gold restorations achieve marginal gaps significantly smaller and more consistent than cast gold. The margin — the thing that matters most — is better.

The workflow advantages are equally significant. From an accurate scan, little to no adjustments are required during seating. Margins are described by dentists as "undetectable" and "spot on." Post-processing is minimal: micro-abrade the intaglio surface, give the exterior a final polish — ten minutes. The restoration is ready to seat.

What was once expensive, difficult, wasteful, and time-consuming is now efficient, affordable, and consistent. The material is the same gold. The delivery system has been modernized.

Chapter 12: The Ceramic Trap

Some dentists, upon hearing the case for gold, respond with a reasonable question: what about ceramic? It is tooth-colored. It is strong. It does not shrink like composite. Surely it solves the margin problem without the aesthetic objection to gold?

It does not. And the manufacturers know it.

Open the Instructions for Use for IPS e.max, Ivoclar Vivadent's market-leading lithium disilicate ceramic. The contraindications state, verbatim: "IPS e.max should not be used on patients with malfunctional occlusion such as bruxers and clenchers." Not "use with caution." Not "consider alternative materials." Should not be used. The manufacturer's own words — an admission that its flagship product is unsuitable for patients who grind or clench. The same contraindication appears in the documentation for zirconia restorations. The companies that design, manufacture, and profit from these materials warn explicitly: do not use this product in patients with parafunctional habits.

Now consider the population. The American Dental Association states that approximately sixty percent of the population will present with some form of parafunction — an imperfect bite, clenching, grinding. Six out of ten patients walking into a dental office. Not a fringe subgroup. The majority. And Ivoclar's own Instructions for Use state that IPS e.max "should not be used on patients with malfunctional occlusion such as bruxers and clenchers." The manufacturer of the most popular ceramic restorative material has declared that six out of ten patients are not candidates for its product.

Yet ceramic restorations are placed in bruxers every day. The contraindication is ignored, the patient is not screened, the restoration is seated, and the opposing dentition begins its slow destruction.

This is not a secret hidden in fine print. Gordon Christensen, the retired dentist and educator who has been zirconia's most prominent advocate — who has spent decades advancing its use through teaching and lecturing — published a report titled "What Dentists and Labs Need to Know in 2020" in the Clinician's Report. He reports a one hundred percent survival rate for the zirconia restorations tested. A respectable result.

But context matters. Gold's benchmark, as we have seen, is ninety-seven percent at thirty years. A material can look flawless at ten and still have a long way to go.

Christensen, to his credit, is honest about zirconia's limitations. The same report lists five clinical situations in which he advises against its use:

Tooth preparations allowing less than 0.6 millimeters of occlusal reduction — the material demands a minimum thickness that many conservative preparations cannot provide.

Zirconia opposing zirconia in heavy occlusion — the forces produce microscopic fractures in the material, which expand over time like a small crack in a windshield. The restoration does not fail suddenly. It degrades progressively, invisibly, until it does.

Zirconia opposing cast gold or polymer — because the hardness differential causes extreme wear. If zirconia wears gold, it will wear enamel.

Where precision attachments are indicated — another solution the profession has largely forgotten but that gold handles effortlessly.

And where optimum aesthetics is a priority, unless the laboratory possesses the narrow skill set to stain zirconia in its green state, fire it correctly, and polish it to the precise point where color emerges without crossing into the gray iridescence that signals over-polishing. But here is the catch: that polished surface is often finished with a glaze — and glaze is soluble in the oral environment. It dissolves. When it does, the abrasive, unpolished zirconia beneath is exposed, and its attrition against the opposing tooth accelerates.

Christensen's five contraindications paint a picture of a material with real limitations — limitations acknowledged by the person who has done more than anyone to promote it.

Ceramic materials — lithium disilicate, zirconia, and their variants — are two to four times harder than natural tooth enamel. They do not wear at the same rate as teeth. When a ceramic crown or onlay opposes a natural tooth, the ceramic wins every time. The opposing tooth loses enamel. Over years of function, the natural dentition is slowly abraded by the restoration that was supposed to protect it.

In a bruxer — the very patient Ivoclar's IFU says should not receive these materials — the forces are higher, the grinding is constant, and the wear is accelerated. The restoration may survive beautifully. The tooth across from it does not.

So the profession's answer to composite's margin failure is a material that survives the margin but destroys what it bites against — and the manufacturers agree this is exactly what will happen. They say do not use it. The profession uses it anyway. The patient trades one problem for another, and the cascade — which began with a single cavity — now involves two teeth.

Gold has no such contraindication. No minimum thickness it demands beyond what conservative preparation provides. No microscopic fracturing under heavy occlusion — gold work-hardens; it does not fracture. No hardness mismatch that destroys opposing teeth — gold wears at the same rate as natural enamel. No precision attachment it cannot accommodate. Its surface does not depend on a soluble glaze that, once gone, leaves behind an abrasive substrate. Gold polishes to a mirror and stays that way.

Gold functions in harmony with the mouth's changing occlusion over decades. A gold inlay and the tooth it opposes age together.

That is not a compromise. That is — acknowledged even by the contraindications of those who sell the alternatives — the clinical ideal for every patient in the practice.

Chapter 13: The Stories That Go Untold

Most gold success stories go unnoticed to the grave, and that is as it should be. A restoration that never fails never makes the news. The patient who received a gold onlay at thirty and died at ninety with the tooth intact never had reason to think about it again. The restoration simply did its job for sixty years and was buried with the patient.

But the profession should notice. Because buried in those silent success stories is the answer to dentistry's most persistent failure.

Caries is the number one disease in the world. Its occurrence rate has climbed for thirty years. Recurrent decay — decay that returns beneath an existing restoration — is the number one reason restorations fail. We are treating cavities, and the cavities are coming back.

The American Dental Association maintains comprehensive caries risk assessment protocols — scoring systems that classify lesions from initial enamel decay to deep cavitations, detection technologies, evidence-based recommendations. The profession has built an elaborate infrastructure around finding decay early.

And when it finds that early lesion — that first cavity, that small shadow on the radiograph, that initial breakdown of enamel — the profession defaults to the material that we know will need retreatment within the next twenty to thirty years. Composite. The least durable material in the operatory.

The irony is complete. We detect early. We assess risk. We document meticulously. And then we fill the cavity with a material that *Streptococcus mutans* consumes, that shrinks as it cures, that fails at the margin, and that will need to be replaced — larger, deeper, more invasively — in seven to ten years. Prevention falls flat when the repair itself is not designed to prevent.

The rise of same-day dentistry has only deepened the contradiction. The materials available for chairside milling — lithium disilicate, resin-ceramic hybrids — are among the least durable in the long term. Same-day delivery solves a scheduling problem for the practice but does not solve the durability problem for the patient. A crown designed, milled, and seated in ninety minutes may feel like progress. A crown that lasts thirty years is progress.

The same-day movement is driven, in part, by a quiet economic pressure: the dental lab bill. For many practices, the monthly invoice from the laboratory is their largest line item after payroll. Eliminating that bill by bringing production in-house looks like savings — and it is, in the short term. But the patient who returns in seven years with recurrent decay beneath that same-day crown is not a saving. The patient who loses the tooth entirely because the restoration failed at the margin is not a saving. A lab bill you can see is a cost. A recurrence you cannot see — until it is too late — is a liability. The profession has been trained to minimize the visible expense and ignore the invisible one.

Gold stops them. Not sometimes. Not in ideal cases. Routinely. Predictably. For decades. This is not luck. This is not operator-dependent. This is inherent.

The evidence is in the mouths of patients walking around right now with gold work placed before their dentist was born. The evidence is in the literature. The evidence is in the margins — closed, sealed, and holding, year after year after year.

Part 4: The Fix

Chapter 14: The Code That Needs to Exist

The CDT code set is maintained by the American Dental Association's Code Maintenance Committee. New codes are proposed, reviewed, and either accepted or rejected through a formal process. It is not a secret process. It is not a conspiracy. It is a bureaucracy — and like most bureaucracies, it responds to sustained, organized pressure.

Currently, there are codes for composite fillings. Codes for amalgam fillings. Codes for ceramic inlays and onlays. Codes for gold crowns — the full-coverage restoration that requires significant tooth reduction. But there is no code for a gold inlay. The most conservative, most durable, most evidence-supported restoration for a small to medium posterior cavity has no billing pathway in the American dental insurance system.

The result of this absence is not neutral. It is directional.

And the insurance contracts reinforce the direction. Here is the actual language from a dental insurance policy:

"Crowns, Inlays, Onlays and Cast Restorations are Benefits only if they are provided to treat cavities which cannot be restored with amalgam, silicate or direct composite (resin) restorations."

Read that again. Gold is positioned not as a first choice, but as a last resort. The contract tells the dentist: try the inferior material first. Only if it is impossible to use amalgam or composite will we consider covering the material that actually works.

The variations across major carriers are consistent in their effect:

The downgrade rule. Many plans downgrade an inlay — regardless of material — to a standard filling allowance. If a gold inlay costs nine hundred dollars, the insurance company calculates its payment based on a one-hundred-and-fifty-dollar composite filling. The patient pays the difference, even with an in-network provider. Gold is covered — at the price of composite.

The tooth-loss threshold. Some contracts require at least fifty percent tooth structure loss before a cast restoration qualifies for coverage. But gold is at its best as a conservative restoration — placed early, preserving healthy tooth structure. The contract requires you to destroy the tooth to save it.

The frequency limit. Inlays and onlays are typically limited to once every five to seven years on the same tooth. This aligns neatly with composite's expected lifespan. It punishes early replacement — and offers no credit for a restoration that lasts decades longer than the limit.

The attrition exclusion. Procedures to repair wear and tear — attrition, abrasion, erosion, bruxism — are explicitly excluded. But wear compatibility is gold's inherent advantage — a property the material simply possesses, requiring no maintenance or intervention. The very thing that makes gold superior is the thing insurance lists as not covered.

Each clause, taken alone, sounds like reasonable cost containment. Taken together, they form a structural preference for the least durable material. The insurance company is not neutral. Its contracts are written to prefer the restoration that fails — because the restoration that fails ensures the patient returns.

All of these guidelines are written to protect the insurance companies for suggesting the least successful material.

Adding a code for gold inlays would not fix every contract overnight. But it would remove the first domino. It would make gold a visible, billable, legitimate option. It would tell the insurance companies: this procedure exists. It would tell the dentists: this procedure is recognized. It would tell the patients: this option is real.

The absence of a code is not a clinical decision. It is a policy failure. And policy failures can be fixed.

Chapter 15: What You Do Tomorrow

If you are a dentist reading this book, you may be wondering what one practitioner can do about a coding system that has been broken for decades. The answer is: more than you think. The CDT Code Maintenance Committee hears from dentists. It responds to submissions. It reviews evidence. The process is designed for exactly this kind of correction — a gap that should not exist, supported by clinical data.

Here is what you can do tomorrow:

First, place a gold inlay. Nothing changes policy faster than procedure. The more gold inlays are placed, coded, and documented — even under a workaround code or as an out-of-pocket procedure — the more the data accumulates. Every gold inlay becomes a data point that a code is needed.

Second, document your outcomes. Margin integrity. Recurrent decay rates. Patient satisfaction. Longevity. The literature already supports gold, but contemporary data from practicing dentists strengthens the submission. A dentist with fifty gold inlays and five years of follow-up data is a more powerful advocate than a dentist with an opinion.

Third, submit to the Code Maintenance Committee. The CDT process accepts proposals from individual practitioners, dental societies, and industry partners. A proposal for a gold inlay code should include the clinical evidence, the gap analysis, and the public health rationale. Template submissions — shared and adapted by dentists across the country — amplify the signal.

Fourth, talk to your colleagues. The coding gap persists partly because most dentists do not know it exists. They were not taught gold. They do not place gold. They do not think about the code that is missing. The first conversation is the hardest. The second is easier. The tenth starts a movement.

Chapter 16: The First Follower

In his TED talk on how movements are started, Derek Sivers shows a video of a lone man dancing at a music festival. He looks ridiculous. People watch and do nothing. Then one person joins him — the first follower — and everything changes. The first follower transforms the lone nut into a leader. The first follower signals to everyone else that it is safe, that it is acceptable, that the dance has begun.

This book is the dance.

I am the lone nut. I have been dancing for years — building Easy Gold Crowns, launching the Gold Dentistry Network, writing, speaking, refining the digital workflow, and telling anyone who will listen that gold is the best first-aid for caries and that the system is broken in ways we can fix.

But I am one person. I need a first follower. And then a second. And then enough of us that the Code Maintenance Committee cannot ignore the submissions piling up on their desks, that the dental schools cannot pretend the material does not exist, that the insurance companies cannot claim there is no demand.

The movement does not require you to abandon composite. It does not require you to stop placing ceramic restorations. It requires you to recognize that gold belongs in the toolkit. That its absence from the codes is not a clinical judgment. That the patient who presents with their first cavity deserves to know every option, including the one that might outlive them.

Join the growing community of dental professionals who want to see the codes change. Submit your evidence. Place your gold. Talk to your colleagues. Be the first follower in your study club, your practice, your town.

The mother wound of restorative dentistry — filling a cavity so that it remains filled, protective, and long-lasting — can be healed. Not by one person. Not by one lab. But by a profession that remembers what it promised.

Epilogue: The Basement

My father has not read this book yet. But he will.

He was president of the Pennsylvania Association of Dental Laboratories when I was still too young to comprehend the gravity of what he was promoting. The binders, the meetings, the drive to bring standards to the profession from the laboratory side. I understand today.

I was eight years old, cutting out construction paper benches, watching our basement fill up with real ones. His work consumed our roller-skating area. A child's perceived loss, but really not.

The mother wound is not just a clinical metaphor. It is personal. It is generational. My father spent his career pushing for higher standards. I have spent mine pushing for a better material. We were both, in our own ways, trying to dress the same wound.

This book is what I found when I followed that wound back to its source — through the codes, the curriculum, the insurance tables, the classroom. And I found something I did not expect: that the fix is within reach. Not easy. Not fast. But possible.

A code on a form. A gold inlay in a molar. A patient who never thinks about that tooth again.

That is the work. It began in a basement in Pennsylvania. It continues with you.